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Intent-Based Networking - Concepts and Definitions

Abstract

Intent and Intent-Based Networking are taking the industry by storm. At the same time, terms related to Intent-Based Networking are often used loosely and inconsistently, in many cases overlapping and confused with other concepts such as "policy." This document clarifies the concept of "intent" and provides an overview of the functionality that is associated with it. The goal is to contribute towards a common and shared understanding of terms, concepts, and functionality that can be used as the foundation to guide further definition of associated research and engineering problems and their solutions.

This document is a product of the IRTF Network Management Research Group (NMRG). It reflects the consensus of the research group, having received many detailed and positive reviews by research group participants. It is published for informational purposes.

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Table of Contents

1. Introduction
2. Definitions and Acronyms
3. Introduction of Concepts
 - 3.1. Intent and Intent-Based Management
 - 3.2. Related Concepts
 - 3.2.1. Service Models
 - 3.2.2. Policy and Policy-Based Network Management
 - 3.2.3. Distinguishing between Intent, Policy, and Service Models
4. Principles
5. Intent-Based Networking - Functionality
 - 5.1. Intent Fulfillment
 - 5.1.1. Intent Ingestion and Interaction with Users
 - 5.1.2. Intent Translation
 - 5.1.3. Intent Orchestration
 - 5.2. Intent Assurance
 - 5.2.1. Monitoring
 - 5.2.2. Intent Compliance Assessment
 - 5.2.3. Intent Compliance Actions
 - 5.2.4. Abstraction, Aggregation, Reporting
6. Life Cycle
7. Additional Considerations

[8. IANA Considerations](#)[9. Security Considerations](#)[10. Informative References](#)[Acknowledgments](#)[Authors' Addresses](#)

1. Introduction

This document is a product of the IRTF Network Management Research Group (NMRG). It reflects the consensus of the RG, receiving reviews and explicit support from many participants. It is published for informational purposes.

In the past, interest regarding management and operations in the IETF has focused on individual network and device features. Standardization emphasis has generally been put on management instrumentation that needed to be provided to a networking device. A prime example of this is SNMP-based management [[RFC3411](#)] and the 200+ MIBs that have been defined by the IETF over the years. More recent examples include YANG data model definitions [[RFC7950](#)] for aspects such as interface configuration, Access Control List (ACL) configuration, and Syslog configuration.

There is a clear sense and reality that managing networks by configuring myriads of "nerd knobs" on a device-by-device basis is no longer an option in modern network environments. Significant challenges arise with keeping device configurations not only consistent across a network but also consistent with the needs of services and service features they are supposed to enable. Additional challenges arise with regard to being able to rapidly adapt the network as needed and to be able to do so at scale. At the same time, operations need to be streamlined and automated wherever possible to not only lower operational expenses but also allow for rapid reconfiguration of networks at sub-second time scales and to ensure that networks are delivering their functionality as expected. Among other things, this requires the ability to consume operational data, perform analytics, and dynamically take actions in a way that is aware of context as well as intended outcomes at near real-time speeds.

Accordingly, the IETF has begun to address end-to-end management aspects that go beyond the realm of individual devices in isolation. Examples include the definition of YANG models for network topology [[RFC8345](#)] or the introduction of service models used by service orchestration systems and controllers [[RFC8309](#)]. Much of the interest has been fueled by the discussion about how to manage autonomic networks as discussed in the ANIMA Working Group. Autonomic networks are driven by the desire to lower operational expenses and make the management of the network as a whole more straightforward, putting it at odds with the need to manage the network one device and one feature at a time. However, while autonomic networks are intended to exhibit "self-management" properties, they still require input from an operator or outside system to provide operational guidance and information about the goals, purposes, and service instances that the network is to serve.

This input and operational guidance are commonly referred to as "intent," and a network that allows network operators to provide their input using intent is referred to as an "Intent-Based Network" (IBN), while a system that helps implement intent is referred to as an "Intent-Based System" (IBS). Those systems can manifest themselves in a number of ways -- for example, as a controller or management system that is implemented as an application that runs on a server or set of servers, or as a set of functions that are distributed across a network and that collectively perform their intent-based functionality.

However, intent is about more than just enabling a form of operator interaction with the network that involves higher-layer abstractions. It is also about the ability to let operators focus on what they want their desired outcomes to be while leaving details to the IBN (respectively IBS) about how those outcomes would be achieved. Focusing on the outcome enables much greater operational efficiency and flexibility at greater scale, in shorter time scales, and with less dependency on human activities (and therefore less possibility for mistakes). This also makes Intent-Based Networking an ideal candidate for artificial intelligence techniques that can bring about the next level of network automation [CLEMM20].

This vision has since caught on with the industry, leading to a significant number of solutions that offer Intent-Based Management that promise network providers to manage networks holistically at a higher level of abstraction and as a system that happens to consist of interconnected components as opposed to a set of independent devices (that happen to be interconnected). Those offerings include IBNs and IBSs (offering a full life cycle of intent), Software-Defined Network (SDN) controllers (offering a single point of control and administration for a network), and network management and Operations Support Systems (OSSs).

It has been recognized for a long time that comprehensive management solutions cannot operate only at the level of individual devices and low-level configurations. In this sense, the vision of intent is not entirely new. In the past, ITU-T's model of a Telecommunications Management Network (TMN) introduced a set of management layers that defined a management hierarchy consisting of network element, network, service, and business management [M3010]. High-level operational objectives would propagate in a top-down fashion from upper to lower layers. The associated abstraction hierarchy was crucial to decompose management complexity into separate areas of concern. This abstraction hierarchy was accompanied by an information hierarchy that concerned itself at the lowest level with device-specific information, but that would, at higher layers, include, for example, end-to-end service instances. Similarly, the concept of Policy-Based Network Management (PBNM) has, for a long time, touted the ability to allow users to manage networks by specifying high-level management policies, with policy systems automatically "rendering" those policies, i.e., breaking them down into low-level configurations and control logic.

As a note, in the context of this document, "users" generally refers to operators and administrators who are responsible for the management and operation of communication services and networking infrastructures, not to "end users" of communication services.

What has been missing, however, is putting these concepts into a more current context and updating them to account for current technology trends. This document clarifies the concepts behind intent. It differentiates intent from related concepts. It also provides an overview of first-order principles of Intent-Based Networking as well as the associated functionality. The goal is to contribute to a common and shared understanding that can be used as a foundation to articulate research and engineering problems in the area of Intent-Based Networking.

It should be noted that the articulation of IBN-related research problems is beyond the scope of this document. However, it should be recognized that Intent-Based Networking has become an important topic in the research community. Per IEEE Xplore [IEEEEXPLORE], as of December 2021, in the past decade since 2012, there have been 1138 papers with the index term "intent", of which 411 specifically mention networking. The time period since 2020 alone accounts for 316 papers on intent and 153 for intent networking, indicating accelerating interest. In addition, workshops dedicated to this theme are beginning to appear, such as the IEEE International Workshop on Intent-Based Networking [WIN21], as well as various special journal issues [IEEE-TITS21]. A survey of current intent-driven networking research has been published in [PANG20], listing among the most pressing current research challenges aspects such as intent translation and understanding, intent interfaces, and security.

2. Definitions and Acronyms

ACL: Access Control List

API: Application Programming Interface

IBA: **Intent-Based Analytics.** Analytics that are defined and derived from users' intent and used to validate the intended state.

IBN: Intent-Based Network. A network that can be managed using intent.

IBS: Intent-Based System. A system that supports management functions that can be guided using intent.

Intent: A set of operational goals (that a network should meet) and outcomes (that a network is supposed to deliver) defined in a declarative manner without specifying how to achieve or implement them.

Intent-Based Management: The concept of performing management based on the concept of intent.

PBNM: Policy-Based Network Management

PDP: Policy Decision Point

PEP: Policy Enforcement Point

Policy: A set of rules that governs the choices in behavior of a system.

Service Model: A model that represents a service that is provided by a network to a user.

SSoT: Single Source of Truth. A functional block in an IBN system that normalizes users' intent and serves as the single source of data for the lower layers.

Statement of Intent: A specification of one particular aspect or component of intent, also referred to as an intent statement.

SVoT: Single Version of Truth

User: In the context of this document, an operator and/or administrator responsible for the management and operation of communication services and networking infrastructure (as opposed to an end user of a communication service).

3. Introduction of Concepts

The following section provides an overview of the concept of intent and Intent-Based Management. It also provides an overview of the related concepts of service models and policies (and Policy-Based Network Management), and explains how they relate to intent and Intent-Based Management.

3.1. Intent and Intent-Based Management

In this document, intent is defined as a set of operational goals (that a network is supposed to meet) and outcomes (that a network is supposed to deliver) defined in a declarative manner without specifying how to achieve or implement them.

The term "intent" was first introduced in the context of Autonomic Networks, where it is defined as "an abstract, high-level policy used to operate the network" [RFC7575]. According to this definition, an intent is a specific type of policy provided by a user to provide guidance to the Autonomic Network that would otherwise operate without human intervention. However, to avoid using intent simply as a synonym for policy, a distinction that differentiates intent clearly from other types of policies needs to be introduced.

One note regarding the use of the plural form of "intent": in the English language, the term "intent" is generally used only in singular form. However, the specification of intent as a whole usually involves the specification of several individual statements of intent, or intent statements. In some cases, intent statements are colloquially also referred to as "intents", although in general, the use of the term "intents" is discouraged.

Intent-Based Management aims to lead towards networks that are fundamentally simpler to manage and operate, requiring only minimal outside intervention. Networks, even when they are autonomic, are not clairvoyant and have no way of automatically knowing particular operational goals nor which instances of networking services to support. In other words, they do not know what the intent of the network provider is that gives the network the purpose of its being. This still needs to be communicated to the network by what informally constitutes intent. That being said, the concept of intent is not limited just to autonomic networks, such as networks that feature an Autonomic Control Plane [RFC8994], but applies to any network.

Intent defines goals and outcomes in a manner that is purely declarative, specifying what to accomplish, not how to achieve it. Intent thus applies several important concepts simultaneously:

- It provides data abstraction: Users do not need to be concerned with low-level device configuration and nerd knobs.
- It provides functional abstraction from particular management and control logic: Users do not need to be concerned even with how to achieve a given intent. What is specified is the desired outcome with the IBS automatically figuring out a course of action (e.g., using an algorithm or applying a set of rules derived from the intent) for how to achieve the outcome.

The following are some examples of intent, expressed in natural language for the sake of clarity (actual interfaces used to convey intent may differ):

- "Steer networking traffic originating from endpoints in one geography away from a second geography, unless the destination lies in that second geography." (states what to achieve, not how.)
- "Avoid routing networking traffic originating from a given set of endpoints (or associated with a given customer) through a particular vendor's equipment, even if this occurs at the expense of reduced service levels." (states what to achieve, not how, providing additional guidance for how to trade off between different goals when necessary.)
- "Maximize network utilization even if it means trading off service levels (such as latency, loss) unless service levels have deteriorated 20% or more from their historical mean." (a desired outcome, with a set of constraints for additional guidance, that does not specify how to achieve this.)
- "Ensure VPN services have path protection at all times for all paths." (a desired outcome of which it may not be clear how it can be precisely accommodated.)
- "Generate in situ Operations, Administration, and Maintenance (OAM) data and network telemetry for later offline analysis whenever significant fluctuations in latency across a path are observed." (goes beyond event-condition-action by not being specific about what constitutes "significant" and what specific data to collect.)
- "Route traffic in a Space Information Network in a way that minimizes dependency on stratospheric balloons unless the intended destination is an aircraft." (does not specify how to precisely achieve this; extrapolates on scenarios mentioned in [\[PANG20\]](#).)
- "For a smart city service, ensure traffic signal control traffic uses dedicated and redundant slices that avoid fate sharing." (a desired outcome with a set of constraints and additional guidance without specifying how to precisely achieve this; extrapolates on scenarios from [\[GHARBAOUI21\]](#).)

In contrast, the following are examples of what would not constitute intent (again, expressed in natural language for the sake of clarity):

- "Configure a given interface with an IP address." (This would be considered device configuration and fiddling with configuration knobs, not intent.)
- "When interface utilization exceeds a specific threshold, emit an alert." (This would be a rule that can help support network automation, but a simple rule is not an intent.)

- "Configure a VPN with a tunnel from A to B over path P." (This would be considered as a configuration of a service.)
- "Deny traffic to prefix P1 unless it is traffic from prefix P2." (This would be an example of an access policy or a firewall rule, not intent.)

In networks, in particular in networks that are deemed autonomic, intent should ideally be rendered by the network itself, i.e., translated into device-specific rules and courses of action. Ideally, intent would not need to be orchestrated or broken down by a higher-level, centralized system but by the network devices themselves using a combination of distributed algorithms and local device abstractions. In this idealized vision, because intent holds for the network as a whole, intent should ideally be automatically disseminated across all devices in the network, which can themselves decide whether they need to act on it.

However, such decentralization will not be practical in all cases. Certain functions will need to be at least conceptually centralized. For example, users may require a single conceptual point of interaction with the network. The system providing this point acts as the operational front end for the network through which users can direct requests at the network and from which they can receive updates about the network. It may appear to users as a single system, even if it is implemented in a distributed manner. In turn, it interacts with and manages other systems in the network as needed in order to realize (i.e., to fulfill and to assure) the desired intent. Likewise, the vast majority of network devices may be intent-agnostic and focus only (for example) on the actual forwarding of packets. Many devices may also be constrained in terms of their processing resources. This means that not every device may be able to act on intent on its own. Again, intent in those cases can be achieved by a separate system that performs the required actions.

Another reason to provide intent functionality from a conceptually centralized point is in cases where the realization of a certain type of intent benefits from global knowledge of a network and its state. In many cases, such a global view may be impractical to maintain by individual devices, for example due to the volume of data and time lags that are involved. It may even be impractical for devices to simply access such a view from another remote system if such were available.

All of this implies that in many cases, certain intent functionality needs to be provided by functions that are specialized for that purpose and that may be provided by dedicated systems (which in some cases could also co-host other networking functions). For example, the translation of specific types of intent into corresponding courses of action and algorithms to achieve the desired outcomes may need to be provided by such specialized functions. Of course, to avoid single points of failure, the implementation and hosting of such functions may still be distributed even if conceptually centralized.

Regardless of its particular implementation in a centralized or decentralized manner, an IBN is a network that can be managed using intent. This means that it is able to recognize and ingest intent of an operator or user and configure and adapt itself according to the user intent, achieving an intended outcome (i.e., a desired state or behavior) without requiring the user to specify the detailed technical steps for how to achieve the outcome. Instead, the IBN will be able to figure out on its own how to achieve the outcome. Similarly, an IBS is a system that allows

users to manage a network using intent. Such a system will serve as a point of interaction with users and implement the functionality that is necessary to achieve the intended outcomes, interacting for that purpose with the network as required.

Other definitions of intent exist, such as [TR523]. Intent there is simply defined as a declarative interface that is typically provided by a controller. It implies the presence of a centralized function that renders the intent into lower-level policies or instructions and orchestrates them across the network. While this is certainly one way of implementation, the definition that is presented here is more encompassing and ambitious, as it emphasizes the importance of managing the network by specifying desired outcomes without the specific steps to be taken in order to achieve the outcome. A controller API that simply provides abstraction at the network level is more limited and would not necessarily qualify as intent. Likewise, ingestion and recognition of intent may not necessarily occur via an API based on function invocations and simple request-response interactions but may involve other types of human-machine interactions such as dialogs to provide clarifications and refinements to requests.

3.2. Related Concepts

3.2.1. Service Models

A service model is a model that represents a service that is provided by a network to a user. Per [RFC8309], a service model describes a service and its parameters in a portable and implementation-agnostic way that can be used independently of the equipment and operating environment on which the service is realized. Two subcategories are distinguished: a "Customer Service Model" describes an instance of a service as provided to a customer, possibly associated with a service order, and a "Service Delivery Model" describes how a service is instantiated over existing networking infrastructure.

An example of a service could be a Layer 3 VPN service [RFC8299], a Network Slice [NETWORK-SLICE], or residential Internet access. Service models represent service instances as entities in their own right. Services have their own parameters, actions, and life cycles. Typically, service instances can be bound to end users of communication services who might be billed for the services provided.

Instantiating a service typically involves multiple aspects:

- A user (or northbound system) needs to define and/or request a service to be instantiated.
- Resources, such as IP addresses, AS numbers, VLAN or VxLAN pools, interfaces, bandwidth, or memory need to be allocated.
- How to map services to the resources needs to be defined. Multiple mappings are often possible, which to select may depend on context (such as which type of access is available to connect the end user of a communication service with the service).
- Bindings between upper- and lower-level objects need to be maintained.
- **Once instantiated, the service operational state needs to be validated and assured to ensure that the network indeed delivers the service as requested.**

The realization of service models involves a system, such as a controller, that provides provisioning logic. This includes breaking down high-level service abstractions into lower-level device abstractions, identifying and allocating system resources, and orchestrating individual provisioning steps. Orchestration operations are generally conducted using a "push" model in which the controller/manager initiates the operations as required, then pushes down the specific configurations to the device and validates whether the new changes have been accepted and the new operational/derived states are achieved and in sync with the intent/desired state. In addition to instantiating and creating new instances of a service, updating, modifying, and decommissioning services also need to be supported. **The device itself typically remains agnostic to the service or the fact that its resources or configurations are part of a service/concept at a higher layer.**

Instantiated service models map to instantiated lower-layer network and device models. Examples include instances of paths or instances of specific port configurations. The service model typically also models dependencies and layering of services over lower-layer networking resources that are used to provide services. This facilitates management by allowing to follow dependencies for troubleshooting activities and to perform impact analysis in which events in the network are assessed regarding their impact on services and customers. Services are typically orchestrated and provisioned top to bottom, which also facilitates keeping track of the assignment of network resources (composition), while troubleshooted bottom up (decomposition). Service models might also be associated with other data that does not concern the network but provides business context. This includes things such as customer data (such as billing information), service orders and service catalogs, tariffs, service contracts, and Service Level Agreements (SLAs), including contractual agreements regarding remediation actions.

[[SERVICE-MAPPING-YANG](#)] is an example of a data model that provides a mapping for customer service models (e.g., the L3VPN Service Model) to Traffic Engineering (TE) models (e.g., the TE Tunnel or the Abstraction and Control of Traffic Engineered Networks Virtual Network model).

Like intent, service models provide higher layers of abstraction. Service models are often also complemented with mappings that capture dependencies between service and device or network configurations. Unlike intent, service models do not allow to define a desired "outcome" that would be automatically maintained by an IBS. Instead, the management of service models requires the development of sophisticated algorithms and control logic by network providers or system integrators.

3.2.2. Policy and Policy-Based Network Management

Policy-Based Network Management (PBNM) is a management paradigm that separates the rules that govern the behavior of a system from the functionality of the system. It promises to reduce maintenance costs of information and communication systems while improving flexibility and runtime adaptability. It is present today at the heart of a multitude of management architectures and paradigms, including SLA-driven, business-driven, autonomous, adaptive, and self-* management [[BOUTABA07](#)]. The interested reader is asked to refer to the rich set of existing literature, which includes this and many other references. In the following, we will only provide a much-abridged and distilled overview.

At the heart of policy-based management is the concept of a policy. Multiple definitions of policy exist: "Policies are rules governing the choices in the behavior of a system" [SLOMAN94]. "Policy is a set of rules that are used to manage and control the changing and/or maintaining of the state of one or more managed objects" [STRASSNER03]. Common to most definitions is the definition of a policy as a "rule." Typically, the definition of a rule consists of an event (whose occurrence triggers a rule), a set of conditions (which get assessed and which must be true before any actions are actually "fired"), and finally, a set of one or more actions that are carried out when the condition holds.

Policy-based management can be considered an imperative management paradigm: Policies precisely specify what needs to be done when and in which circumstance. By using policies, management can, in effect, be defined as a set of simple control loops. This makes policy-based management a suitable technology to implement autonomic behavior that can exhibit self-* management properties, including self-configuration, self-healing, self-optimization, and self-protection. This is notwithstanding the fact that policy-based management may make use of the concept of abstractions (such as, "Bob gets gold service") that hide from the user the specifics of how that abstraction is rendered in a particular deployment.

Policies typically involve a certain degree of abstraction in order to cope with the heterogeneity of networking devices. Rather than having a device-specific policy that defines events, conditions, and actions in terms of device-specific commands, parameters, and data models, a policy is defined at a higher level of abstraction involving a canonical model of systems and devices to which the policy is to be applied. A policy agent on a controller or the device subsequently "renders" the policy, i.e., translates the canonical model into a device-specific representation. This concept allows applying the same policy across a wide range of devices without needing to define multiple variants. In other words, policy definition is decoupled from policy instantiation and policy enforcement. This enables operational scale and allows network operators and authors of policies to think in higher terms of abstraction than device specifics and be able to reuse the same, high-level definition across different networking domains, WAN, data center (DC), or public cloud.

PBNM is typically "push-based": Policies are pushed onto devices where they are rendered and enforced. The push operations are conducted by a manager or controller that is responsible for deploying policies across the network and monitoring their proper operation. That being said, other policy architectures are possible. For example, policy-based management can also include a pull component in which the decision regarding which action to take is delegated to a so-called Policy Decision Point (PDP). This PDP can reside outside the managed device itself and has typically global visibility and context with which to make policy decisions. Whenever a network device observes an event that is associated with a policy but lacks the full definition of the policy or the ability to reach a conclusion regarding the expected action, it reaches out to the PDP for a decision (reached, for example, by deciding on an action based on various conditions). Subsequently, the device carries out the decision as returned by the PDP; the device "enforces" the policy and hence acts as a PEP (Policy Enforcement Point). Either way, PBNM architectures typically involve a central component from which policies are deployed across the network and/or policy decisions served.

Like intent, policies provide a higher layer of abstraction. Policy systems are also able to capture dynamic aspects of the system under management through the specification of rules that allow defining various triggers for specific courses of action. Unlike intent, the definition of those rules (and courses of action) still needs to be articulated by users. Since the intent is unknown, conflict resolution within or between policies requires interactions with a user or some kind of logic that resides outside of PBNM. In that sense, policy constitutes a lower level of abstraction than intent, and it is conceivable for IBSs to generate policies that are subsequently deployed by a PBNM system, allowing PBNM to support Intent-Based Networking.

3.2.3. Distinguishing between Intent, Policy, and Service Models

What intent, policy, and service models all have in common is the fact that they involve a higher layer of abstraction of a network that does not involve device specifics, generally transcends individual devices, and makes the network easier to manage for applications and human users compared to having to manage the network one device at a time. Beyond that, differences emerge.

Summarized differences:

- A service model is a data model that is used to describe instances of services that are provided to customers. A service model has dependencies on lower-level models (device and network models) when describing how the service is mapped onto the underlying network and IT infrastructure. Instantiating a service model requires orchestration by a system; the logic for how to orchestrate/manage/provide the service model and how to map it onto underlying resources is not included as part of the model itself.
- Policy is a set of rules, typically modeled around a variation of events/conditions/actions, used to express simple control loops that can be rendered by devices without requiring intervention by the outside system. Policies let users define what to do under what circumstances, but they do not specify the desired outcome.
- Intent is a high-level, declarative goal that operates at the level of a network and services it provides, not individual devices. It is used to define outcomes and high-level operational goals, without specifying how those outcomes should be achieved or how goals should specifically be satisfied, and without the need to enumerate specific events, conditions, and actions. Which algorithm or rules to apply can be automatically "learned/derived from intent" by the IBS. In the context of autonomic networking, intent is ideally rendered by the network itself; also, the dissemination of intent across the network and any required coordination between nodes is resolved by the network without the need for external systems.

One analogy to capture the difference between policy-based systems and IBSs is that of Expert Systems and Learning Systems in the field of Artificial Intelligence. Expert Systems operate on knowledge bases with rules that are supplied by users, analogous to policy systems whose policies are supplied by users. They are able to make automatic inferences based on those rules but are not able to "learn" new rules on their own. Learning Systems (popularized by deep learning and neural networks), on the other hand, are able to learn without depending on user programming or articulation of rules. However, they do require a learning or training phase

requiring large data sets; explanations of actions that the system actually takes provide a different set of challenges. Analogous to IBSs, users focus on what they would like the learning system to accomplish but not how to do it.

4. Principles

The following main operating principles allow characterizing the intent-based/-driven/-defined nature of a system.

1. Single Source of Truth (SSoT) and Single Version of Truth (SVoT). The SSoT is an essential component of an IBS as it enables several important operations. The set of validated intent expressions is the system's SSoT. SSoT and the records of the operational states enable comparing the intended/desired state and actual/operational states of the system and determining drift between them. SSoT and the drift information provide the basis for corrective actions. If the IBS is equipped with the means to predict states, it can further develop strategies to anticipate, plan, and pro-actively act on any diverging trends with the aim to minimize their impact. Beyond providing a means for consistent system operation, SSoT also allows for better traceability to validate if/how the initial intent and associated business goals have been properly met in order to evaluate the impacts of changes in the intent parameters and impacts and effects of the events occurring in the system.

Single Version (or View) of Truth derives from the SSoT and can be used to perform other operations such as querying, polling, or filtering measured and correlated information in order to create so-called "views." These views can serve the users of the IBS. In order to create intent statements as single sources of truth, the IBS must follow well-specified and well-documented processes and models. In other contexts, SSoT is also referred to as the invariance of the intent [LENROW15].

2. One touch but not one shot. In an ideal IBS, the user expresses intent in one form or another, and then the system takes over all subsequent operations (one touch). A zero-touch approach could also be imagined in the case where the IBS has the capabilities or means to recognize intentions in any form of data. However, the zero- or one-touch approach should not distract from the fact that reaching the state of a well-formed and valid intent expression is not a one-shot process. On the contrary, the interfacing between the user and the IBS could be designed as an interactive and iterative process. Depending on the level of abstraction, the intent expressions may initially contain more or less implicit parts and imprecise or unknown parameters and constraints. The role of the IBS is to parse, understand, and refine the intent expression to reach a well-formed and valid intent expression that can be further used by the system for the fulfillment and assurance operations. An intent refinement process could use a combination of iterative steps involving the user to validate the proposed refined intent and to ask the user for clarifications in case some parameters or variables could not be deduced or learned by means of the system itself. In addition, the IBS will need to moderate between conflicting intent, helping users to properly choose between intent alternatives that may have different ramifications.
3. Autonomy and Supervision. A desirable goal for an IBS is to offer a high degree of flexibility and freedom on both the user side and system side, e.g., by giving the user the ability to express intent using the user's own terms, by supporting different forms of expression for

individual statements of intent and being capable of refining the intent expressions to well-formed and exploitable expressions. The dual principle of autonomy and supervision allows operating a system that will have the necessary levels of autonomy to conduct its tasks and operations without requiring the intervention of the user and taking its own decisions (within its areas of concern and span of control) as how to perform and meet the user expectations in terms of performance and quality, while at the same time providing the proper level of supervision to satisfy the user requirements for reporting and escalation of relevant information.

4. **Learning. An IBS is a learning system.** In contrast to an imperative type of system, such as Event-Condition-Action policy rules, where the user defines beforehand the expected behavior of the system to various events and conditions, in an IBS, the user only declares what the system is supposed to achieve and not how to achieve these goals. **There is thus a transfer of reasoning/rationality from the human (domain knowledge) to the system.** This transfer of cognitive capability also implies the availability in the IBS of capabilities or means for learning, reasoning, and knowledge representation and management. **The learning abilities of an IBS can apply to different tasks such as optimization of the intent rendering or intent refinement processes.** **The fact that an IBS is a continuously evolving system creates the condition for continuous learning and optimization.** Other cognitive capabilities such as planning can also be leveraged in an IBS to anticipate or forecast future system state and response to changes in intent or network conditions and thus elaboration of plans to accommodate the changes while preserving system stability and efficiency in a trade-off with cost and robustness of operations.
5. **Capability exposure.** Capability exposure consists in the need for expressive network capabilities, requirements, and constraints to be able to compose/decompose intent and map the user's expectations to the system capabilities.
6. **Abstract and outcome-driven.** Users do not need to be concerned with how intent is achieved and are empowered to think in terms of outcomes. In addition, they can refer to concepts at a higher level of abstractions, independent, e.g., of vendor-specific renderings.

The described principles are perhaps the most prominent, but they are not an exhaustive list. There are additional aspects to consider, such as:

- Intent targets are not individual devices but typically aggregations (such as groups of devices adhering to a common criteria, such as devices of a particular role) or abstractions (such as service types, service instances, or topologies).
- Abstraction and inherent virtualization: agnostic to implementation details.
- Human-tailored network interaction: IBN should speak the language of the user as opposed to requiring the user to speak the language of the device/network.
- Explainability as an important IBN function, detection and IBN-aided resolution of conflicting intent, **and reconciliation of what the user wants and what the network can actually do.**
- Inherent support, verification, and assurance of trust.

All of these principles and considerations have implications on the design of IBSs and their supporting architecture. Accordingly, they need to be considered when deriving functional and operational requirements.

5. Intent-Based Networking - Functionality

Intent-Based Networking involves a wide variety of functions that can be roughly divided into two categories:

- Intent Fulfillment provides functions and interfaces that allow users to communicate intent to the network and that perform the necessary actions to ensure that intent is achieved. This includes **algorithms to determine proper courses of action** and **functions that learn to optimize outcomes over time**. In addition, it also includes functions such as any required **orchestration of coordinated configuration operations across the network and rendering of higher-level abstractions into lower-level parameters and control knobs**.
- Intent Assurance provides functions and interfaces that **allow users to validate and monitor that the network is indeed adhering to and complying with intent**. This is necessary to assess the effectiveness of actions taken as part of fulfillment, **providing important feedback that allows those functions to be trained or tuned over time to optimize outcomes**. In addition, Intent Assurance is necessary to address "intent drift." Intent is not meant to be transactional, i.e., "set and forget", but is expected to remain in effect over time (unless explicitly stated otherwise). Intent drift occurs when a system originally meets the intent, but over time gradually allows its behavior to change or be affected until it no longer does or does so in a less effective manner.

The following sections provide a more comprehensive overview of those functions.

5.1. Intent Fulfillment

Intent fulfillment is concerned with the functions that take intent from its origination by a user (generally, an administrator of the responsible organization) to its realization in the network.

5.1.1. Intent Ingestion and Interaction with Users

The first set of functions is concerned with "ingesting" intent, i.e., obtaining intent through interactions with users. They provide functions that recognize intent from interaction with the user as well as functions that allow users to refine their intent and articulate it in such ways so that it becomes actionable by an IBS. Typically, those functions go beyond those provided by a non-intent-based API, although non-intent-based APIs may also still be provided (and needed for interactions beyond human users, i.e., with other machines). Many cases would also involve a set of intuitive and easy-to-navigate workflows that guide users through the intent ingestion phase, making sure that all inputs that are necessary for intent modeling and consecutive translation have been gathered. **They may support unconventional human-machine interactions, in which a human will not simply give commands but instead a human-machine dialog is used to provide clarifications, to explain ramifications and trade-offs, and to facilitate refinements.**

The goal is ultimately to make IBSs as easy and natural to use and interact with as possible, in particular allowing human users to interact with the IBS in ways that do not involve a steep learning curve that forces the user to learn the "language" of the system. Ideally, it will be the IBSs that are increasingly able to learn how to understand the user, as opposed to the other way around. **Of course, further research will be required to make this a reality.**

5.1.2. Intent Translation

A second set of functions needs to **translate user intent into courses of action and requests to take against the network, which will be meaningful to network configuration and provisioning systems.** These functions lie at the core of IBS, bridging the gap between interaction with users on the one hand and the management and operations side that will need to orchestrate provisioning and configuration across the network.

Beyond merely breaking down a higher layer of abstraction (intent) into a lower layer of abstraction (policies and device configuration), **Intent Translation functions can be complemented with functions and algorithms that perform optimizations and that are able to learn and improve over time in order to result in the best outcomes, specifically in cases where multiple ways of achieving those outcomes are conceivable.** For example, satisfying an intent may involve computation of paths and other parameters that will need to be configured across the network. Heuristics and algorithms to do so may evolve over time to optimize outcomes that may depend on a myriad of dynamic network conditions and context.

5.1.3. Intent Orchestration

A third set of functions deals with the actual configuration and provisioning steps that need to be orchestrated across the network and that were determined by the previous intent translation step.

5.2. Intent Assurance

Intent Assurance is concerned with the functions that are necessary to ensure that the network indeed complies with the desired intent once it has been fulfilled.

5.2.1. Monitoring

A first set of assurance functions monitors and observes the network and its exhibited behavior. This includes all the usual assurance functions such as **monitoring the network for events and performance outliers, performing measurements to assess service levels that are being delivered, and generating and collecting telemetry data.** Monitoring and observation are required as the basis for the next set of functions that **assess whether the observed behavior is in fact in compliance with the behavior that is expected based on the intent.**

5.2.2. Intent Compliance Assessment

At the core of Intent Assurance are functions that **compare the actual network behavior that is being monitored and observed with the intended behavior that is expected per the intent and is held by SSoT.** These functions continuously assess and validate whether the observation indicates compliance with intent. This includes assessing the effectiveness of intent fulfillment

actions, including verifying that the actions had the desired effect and assessing the magnitude of the effect as applicable. **It can also include functions that perform analysis and aggregation of raw observation data. The results of the assessment can be fed back to facilitate learning functions that optimize outcomes.**

Intent compliance assessment also includes assessing whether intent drift occurs over time. Intent drift can be caused by a control plane or lower-level management operations that inadvertently cause behavior changes that conflict with intent that was orchestrated earlier. IBSs and Networks need to be able to detect when such drift occurs or is about to occur as well as assess the severity of the drift.

5.2.3. Intent Compliance Actions

When intent drift occurs or network behavior is inconsistent with desired intent, functions that are able to trigger corrective actions are needed. This includes actions needed to resolve intent drift and bring the network back into compliance. Alternatively, and where necessary, reporting functions need to be triggered **that alert operators and provide them with the necessary information and tools to react appropriately, e.g., by helping them articulate modifications to the original intent to moderate between conflicting concerns.**

5.2.4. Abstraction, Aggregation, Reporting

The outcome of Intent Assurance needs to be reported back to the user in ways that allow the user to relate the outcomes to their intent. This requires **a set of functions that are able to analyze, aggregate, and abstract the results of the observations accordingly. In many cases, lower-level concepts such as detailed performance statistics and observations related to low-level settings need to be "up-leveled" to concepts the user can relate to and take action on.**

The required aggregation and analysis functionality needs to be complemented with functions that report intent compliance status and provide adequate summarization and visualization to human users.

6. Life Cycle

Intent is subject to a life cycle: it comes into being, may undergo changes over the course of time, and may at some point be retracted. This life cycle is closely tied to various interconnection functions that are associated with the intent concept.

Figure 1 depicts an intent life cycle and its main functions. The functions were introduced in Section 5 and are divided into two functional (horizontal) planes reflecting the distinction between fulfillment and assurance. In addition, they are divided into three (vertical) spaces.

The spaces indicate the different perspectives and interactions with different roles that are involved in addressing the functions:

- The User Space involves the functions that interface the network and IBS with the human user. It involves the functions that allow users to articulate and the IBS to recognize that intent. It also involves the functions that report back the status of the network relative to the

intent and that allow users to assess outcomes and whether their intent has the desired effect.

- The Translation, or Intent-Based System (IBS) Space involves the functions that bridge the gap between intent users and the network operations infrastructure. This includes the functions used to translate an intent into a course of action as well as the algorithms that are used to plan and optimize those courses of action also in consideration of feedback and observations from the network. It also includes the functions to analyze and aggregate observations from the network in order to validate compliance with the intent and to take corrective actions as necessary. **In addition, it includes functions that abstract observations from the network in ways that relate them to the intent as communicated by users.** This facilitates the reporting functions in the user space.
- The Network Operations Space, finally, involves orchestration, configuration, monitoring, and measurement functions, which are used to effectuate the rendered intent and observe its effects on the network.

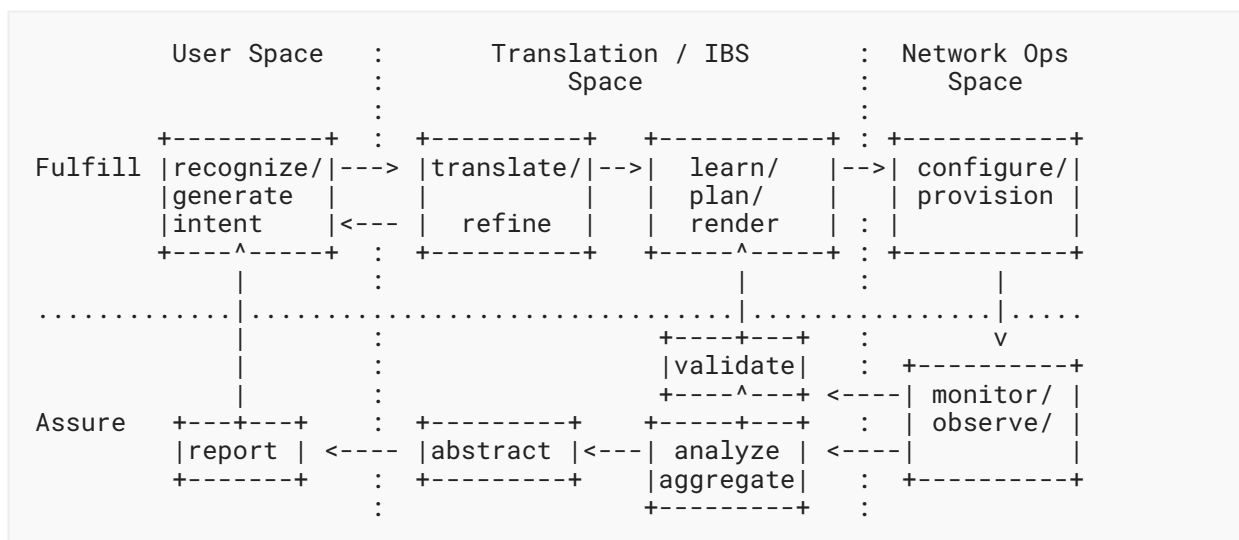


Figure 1: Intent Life Cycle

When carefully inspecting the diagram, it becomes apparent that the intent life cycle, in fact, involves two cycles, or loops:

- **The "inner" intent control loop between IBS and Network Operations space is completely autonomic and does not involve any human in the loop. It represents closed-loop automation that involves automatic analysis and validation of intent based on observations from the network operations space. Those observations are fed into the function that plans the rendering of networking intent in order to make adjustments as needed in the configuration of the network. The loop addresses and counteracts any intent drift that may be occurring, using observations to assess the degree of the network's intent compliance and automatically prompting actions to address any discrepancies. Likewise, the loop allows to assess the effectiveness of any actions that are taken in order to continuously learn and improve how intent needs to be rendered in order to achieve the desired outcomes.**

- The "outer" intent control loop extends to the user space. It includes the user taking action and adjusting their intent based on observations and feedback from the IBS. Intent is thus subjected to a life cycle: It comes into being, may undergo refinements, modifications, and changes of time, and may at some point in time even get retracted.

7. Additional Considerations

Given the popularity of the term "intent," it is tempting to broaden its use to encompass other related concepts, resulting in "intent-washing" that paints those concepts in a new light by simply applying new intent terminology to them. A common example concerns referring to the northbound interface of SDN controllers as "intent interface." However, in some cases, this actually makes sense not just as a marketing ploy but as a way to better relate previously existing and new concepts.

In that sense and with regards to intent, it makes sense to distinguish various subcategories of intent as follows:

Operational Intent: defines intent related to operational goals of an operator; it corresponds to the original "intent" term and the concepts defined in this document.

Rule Intent: a synonym for policy rules regarding what to do when certain events occur.

Service Intent: a synonym for customer service model [[RFC8309](#)].

Flow Intent: a synonym for a Service Level Objective for a given flow.

A comprehensive set of classifications of different concepts and categories of intent will be described in a separate document.

8. IANA Considerations

This document has no IANA actions.

9. Security Considerations

This document describes concepts and definitions of Intent-Based Networking. As such, the below security considerations remain high level, i.e., in the form of principles, guidelines, or requirements. More detailed security considerations will be described in the documents that specify the architecture and functionality.

Security in Intent-Based Networking can apply to different facets:

- Securing the IBS itself.
- Mitigating the effects of erroneous, harmful, or compromised intent statements.
- Expressing security policies or security-related parameters with intent statements.

Securing the IBS aims at making the IBS operationally secure by implementing security mechanisms and applying security best practices. In the context of Intent-Based Networking, such mechanisms and practices may consist of **intent verification and validation, operations on intent by authenticated and authorized users only, and protection against or detection of tampered statements of intent**. Such mechanisms may also include the introduction of multiple levels of intent. **For example, intent related to securing the network should occur at a "deeper" level that overrides other levels of intent if necessary**, and that is not subject to modification through regular operations but through ones that are specifically secured. Use of additional mechanisms such as explanation components that describe the security ramifications and trade-offs should be considered as well.

Mitigating the effects of erroneous or compromised statements of intent aims at making the IBS operationally safe by **providing checkpoint and safeguard mechanisms and operating principles**. In the context of Intent-Based Networking, such mechanisms and principles may consist of **the ability to automatically detect unintended, detrimental, or abnormal behavior; the ability to automatically (and gracefully) roll back or fall back to a previous "safe" state**; the ability to prevent or contain error amplification (due to the combination of a higher degree of automation and the intrinsic higher degree of freedom, ambiguity, and implicit information conveyed by intent statements); and dynamic levels of supervision and reporting to make the user aware of the right information at the right time with the right level of context. Erroneous or harmful intent statements may inadvertently propagate and compromise security. In addition, compromised intent statements (for example, forged by an inside attacker) may sabotage or harm the network resources and make them vulnerable to further, larger attacks, e.g., by defeating certain security mechanisms.

Expressing security policies or security-related parameters as intent consists of **using the intent formalism (a high-level, declarative abstraction) or part(s) of an intent statement to define security-related aspects** such as:

- data governance;
- level(s) of confidentiality in data exchange;
- level(s) of availability of system resources, of protection in forwarding paths, and of isolation in processing functions;
- level(s) of encryption; and
- authorized entities for given operations.

The development and introduction of Intent-Based Networking in operational environments **will certainly create new security concerns**. Such security concerns have to be anticipated at the design and specification time. **However, Intent-Based Networking may also be used as an enabler for better security. For instance, security and privacy rules could be expressed in a more human-friendly and generic way and be less technology specific and less complex, leading to fewer low-level configuration mistakes. The detection of threats or attacks could also be made more simple and comprehensive thanks to conflict detection at higher level or at coarser granularity.**

More thorough security analyses should be conducted as our understanding of Intent-Based Networking technology matures.

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